THINK LIKE A SCIENTIST 像科学家一样思考 · QUICK TEACHER GUIDE · STANDALONE PRACTICAL · 30 MINUTES HANDS-ON

DNA from fruit 从水果中提取DNA

Squash the fruit in soapy, salty water, filter it, pour ice-cold alcohol on top, wait: white strands of DNA appear at the line between the layers.

KIT PER GROUP (3 OR 4 STUDENTS)

- **Half a ripe kiwi, peeled** (or a third of a ripe banana; strawberries work best of all)
- **20 ml extraction liquid** (4 teaspoons)
- Zip-lock bag (or cup and fork)
- Coffee filter (or 2 sheets of kitchen paper, J-cloth, tea strainer) in a funnel (or cut-off bottle top) over a clear cup
- Test tube (or any narrow clear tube or small bottle: narrow shows the layers best)
- **About 10 ml ice-cold alcohol**, handed out by you at step 6
- Wooden skewer or cocktail stick; teaspoon or 10 ml syringe (no needle); goggles each

Class of 8 groups: 4 kiwis plus 1 spare · 200 ml extraction liquid · about 150 ml alcohol.

EXTRACTION LIQUID (8 GROUPS, MAKE ON THE DAY)

180 ml water (tap is fine) + **20 ml washing-up liquid** (4 tsp) + **6 g salt** (1 level tsp). Stir gently, do not whisk: you want little froth. Label: Extraction liquid 提取液.

THE ALCOHOL: STRENGTH MATTERS MORE THAN TYPE

95% ethanol 95%酒精 **or anhydrous** 无水乙醇	**Best.** Used in the Chinese senior-school practical DNA的粗提取与鉴定: ask the prep room first
Isopropyl 90%+ 异丙醇	**Very good**
Methylated spirits / denatured	**Works.** Purple dye does not stop it
75% medical alcohol 75%医用酒精**, 70% rubbing alcohol**	**Works, gives less.** Very cold, only 2 cm of fruit liquid, deeper alcohol layer, wait the full 5 min. **Trial run first**
Hand gel, glycerine, perfume, wipes, iodine 碘伏**, drinking spirits**	**No**

Chill it: 150 ml in a small, capped, labelled plastic bottle in the freezer for a few hours (it will not freeze; keep the cap tight, short time only). No freezer: bowl of ice water for 30 min. In the lesson, stand it in a cup of ice.

IF IT DOES NOT WORK

Nothing appears	Wait 5 to 10 min; trickle on more cold alcohol. Next time: stronger, colder, less liquid in the tube
Cloudy all through	Poured too fast or shaken. Leave it still
Thick froth	Squashed too hard. Leave the froth behind
Pulp in the tube	Filter was squeezed. Let it settle, pour off
Banana is gloopy	Starchy fruit. Use kiwi, or less banana

METHOD AND WHAT TO SAY

MIN	DO / say (the why)
0 to 5	**Goggles on, safety briefing.** Hook: "Is there DNA in your breakfast?" Fruit is made of cells; every cell has DNA, the instructions for building it.
5 to 9	**1 to 3. Fruit + 20 ml liquid in the bag, air out, seal, squash gently 2 to 3 min.** No big lumps, little froth. Squashing breaks open the cell walls.
9 to 17	**4. Pour into the filter, let it drip 5 min. Do not squeeze.** Washing-up liquid breaks the oily membranes round the cell and nucleus (like grease on plates), so DNA comes out. Salt helps the strands clump. The filter removes solids: the DNA is in the liquid.
17 to 19	**5. Filtered liquid into the tube, a quarter full (2 to 3 cm).**
19 to 21	**6 to 7. You hand out the alcohol. Tilt the tube, trickle it slowly down the side until the alcohol layer is twice as deep as the fruit layer. No shaking.** Alcohol floats: it is less dense than water.
21 to 26	**8. Hold still, watch the line for 3 to 5 min.** DNA does not dissolve in alcohol, so it comes out as white strands where the layers meet. Cold gives more and slows the fruit's enzymes that chop DNA up.
26 to 31	**9. Stick in at the line, twist slowly like spaghetti, lift out.**
31 to 40	**Look, explain, clean up.**

WHAT THEY ARE LOOKING AT

- **Millions of DNA strands clumped together**, with some protein and fruit pectin. A rough extract, not pure DNA.
- **Not the double helix:** one molecule is about 2 nm wide, far too thin to see.
- Kiwi cells have 6 copies of each chromosome set, banana 3, strawberry 8: more copies, more DNA.
- The DNA in one human cell, stretched out, is about 2 m long.
- Watch for: "the alcohol makes the DNA" (no, it was already there); "fruit is not alive so has no DNA"; "DNA is only in the seeds".

SAFETY 注意安全

- **Flammable:** no flames or heaters; you hand out small amounts; lid on; window open. 酒精易燃，附近不能有明火。
- **Poisonous:** never drink or taste; nothing here is eaten. 不要喝，也不要尝。
- **Eyes:** goggles; splash: rinse 10 min with water, tell the teacher. 请戴护目镜。
- **Allergy:** check for kiwi, banana and latex; allergic students swap fruit or watch; no latex gloves. 课前确认过敏情况。
- **Spills:** wipe up with paper towel at once, open a window.

PREP

Trial run with the exact alcohol you bought, a few days before (15 min).

A FEW DAYS BEFORE

- [ ] Allergy list (kiwi, banana, latex) · get alcohol · buy ripe fruit (kiwi soft when pressed)

THE DAY BEFORE

- [ ] Alcohol in freezer, labelled: Cold alcohol: flammable, do not drink 冷酒精：易燃，不可饮用

ON THE DAY, ABOUT 15 MIN

- [ ] Extraction liquid · peel and halve fruit · a tray per group · cup of ice on your desk

CLEAN-UP

- [ ] Mush and filters bagged in the bin · used liquids down the sink with lots of water (check school rules) · spare alcohol in the flammables cupboard, not back in the stock bottle · wash hands · kept DNA stays in class

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Student step card 学生步骤卡 · one per group 每组一张

Goggles on. Alcohol is flammable and poisonous: never drink or taste it. 戴上护目镜。酒精易燃、有毒：不要喝，也不要尝。

1. Put on your goggles.
戴上护目镜。
2. Put the fruit and **4 teaspoons** of extraction liquid in the bag. Squeeze out the air and seal it.
把水果和4茶匙提取液放进袋子里。挤出空气，封好袋子。
3. Gently squash the fruit for **2 minutes**. Try not to make bubbles.
轻轻挤压水果2分钟。尽量不要产生泡沫。
4. Pour it into the filter. Let it drip for **5 minutes**. Do not squeeze.
把它倒进滤纸里。让它滴5分钟。不要挤压。
5. Pour the liquid into the tube until it is **a quarter full**.
把滤出的液体倒进试管，到四分之一满。
6. Ask your teacher for the cold alcohol.
向老师领取冷酒精。
7. Tilt the tube. Pour the alcohol **slowly down the side**. Do not shake.
把试管倾斜。沿着管壁慢慢倒入酒精。不要摇晃。
8. Wait **3 to 5 minutes**. Watch the line between the two layers.
等3到5分钟。观察两层液体之间的分界线。
9. Twist the stick at the line and lift out the white strands. **This is DNA!**
在分界线处慢慢转动小棍，把白色的丝挑出来。这就是DNA！
10. Clean up and wash your hands.
清理桌面，洗手。

THINK 想一想

1. Why do we squash the fruit? 为什么要挤压水果?
2. What does the washing-up liquid do? 洗洁精有什么作用?
3. Why does the DNA appear where the alcohol is? 为什么DNA出现在酒精层?

✂ cut here

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